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Abstract

This project presents the design of a digital firing circuit for a single-phase inverter using an 8051 microcontroller to generate a 50Hz square wave for triggering MOSFETs. The MOSFET outputs, powered by a 12V DC supply, are combined using an op-amp subtractor to form a single 12V square wave, which is then converted to an 11V sine wave via a 4th-order low-pass filter.

The circuit was simulated in Proteus and the PCB design was created in KiCAD, with successful verification through ERC and DRC checks. Simulation results and PCB layouts confirm the system's functionality in generating a sine wave output suitable for AC applications.

Chapter 1

Introduction

1.1 Background

Inverters are crucial components in power electronics, converting DC (Direct Current) power into AC (Alternating Current) power, which is used to run many household appliances, industrial equipment, and renewable energy systems. A single-phase inverter is commonly utilized in smaller-scale applications such as solar power systems and uninterruptible power supplies (UPS). These systems require precise control over the switching of MOSFETs to generate the appropriate AC output, typically in the form of a sine wave.

Traditionally, analog control methods have been employed for such tasks. However, with advancements in digital electronics, microcontrollers offer a more flexible, programmable, and accurate way to manage the switching process in these circuits. This project focuses on using an 8051 microcontroller to create a digital firing circuit that controls the MOSFETs in a single-phase inverter, converting DC input into a stable AC output.

1.2 Motivation

The primary motivation behind this project is to explore the use of micro-controller based digital control in inverter systems. The 8051 micro-controller provides precise timing and control over the MOSFET switching, which is essential for producing the desired output waveform in a single-phase inverter. Additionally, this project aims to demonstrate the process of converting a square wave into a smooth sine wave using a low-pass filter, a key requirement in reducing harmonic distortion in AC power systems.

Moreover, the project offers practical experience in the simulation of power electronics circuits using Proteus and in the design of PCB layouts with KiCAD. The ability to simulate, design, and implement a digital control system for an inverter opens up further opportunities to optimize and innovate in power electronics applications.

Chapter 2

Methodology

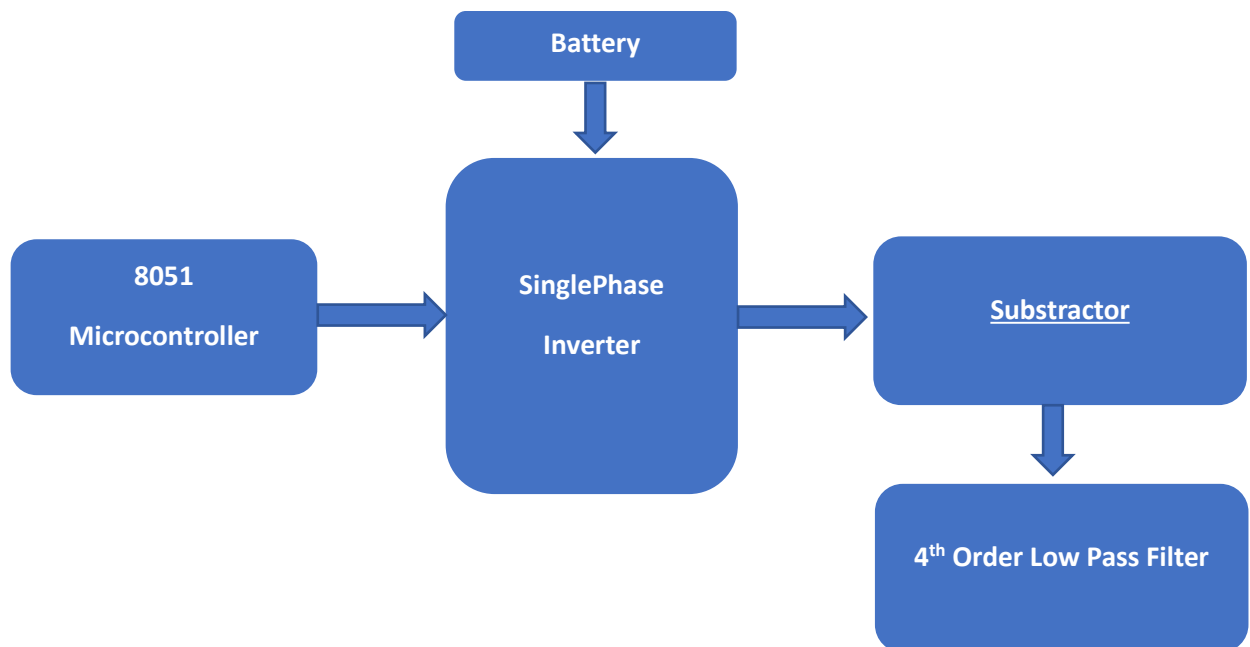
1. **Designing the Digital Firing Circuit:** The project begins by designing a digital firing circuit using the 8051 micro-controller. The micro-controller is programmed to generate two inverted 50Hz square waves for triggering the MOSFETs, ensuring precise control of the switching process.
2. **MOSFET Switching and Power Conversion:** The 8051 triggers two MOSFETs, powered by a 12V DC supply. The complementary signals from the MOSFETs are fed into an op-amp subtractor, which combines them into a single 12V square wave, forming the base for AC power conversion.
3. **Waveform Smoothing:** The square wave output is passed through a 4th-order low-pass filter, designed using op-amps, to smooth the signal into an 11V sine wave. This is essential for producing a stable AC output with minimal harmonic distortion, suitable for AC-powered applications.
4. **Circuit Simulation:** The entire circuit is simulated in Proteus software. The simulation verifies the correct behavior of the firing circuit, the MOSFET operation, and the generation of both square and sine waveforms, ensuring the circuit's functionality before physical implementation.
5. **PCB Design and Verification:** Using KiCAD, the schematic of the circuit is recreated and the components are assigned appropriate footprints. Electrical Rule Check (ERC) is run to ensure logical consistency, followed by Design Rule Check (DRC) to verify compliance with PCB manufacturing rules.
6. **PCB Layout Creation:** After validating the schematic, the PCB layout is designed, focusing on optimizing trace routing and ensuring proper footprint dimensions. The final design is checked for manufacturability and signal integrity.

Chapter 3

Design & Implementation

3.1 Block Diagram

The system design consists of several key components working together to generate a sine wave output from a DC input source.



3.2 Components Used

1. **8051 Microcontroller:** This is the central unit responsible for generating two complementary 50Hz square wave signals to trigger the MOSFETs. These signals are programmed and controlled using a timer interrupt within the microcontroller.



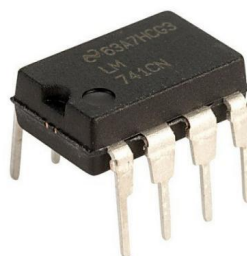
2. **MOSFETs (IRF540):** The IRF540 MOSFETs act as electronic switches, converting the DC input from the 12V battery into square wave outputs. They are driven by the square waves generated by the 8051 microcontroller, and the switching alternates between two MOSFETs, creating complementary signals.



3. **Transistor (BC547):** The BC547 NPN transistor is used to further control the MOSFET gates, providing the necessary drive current and isolation between the microcontroller and MOSFETs.



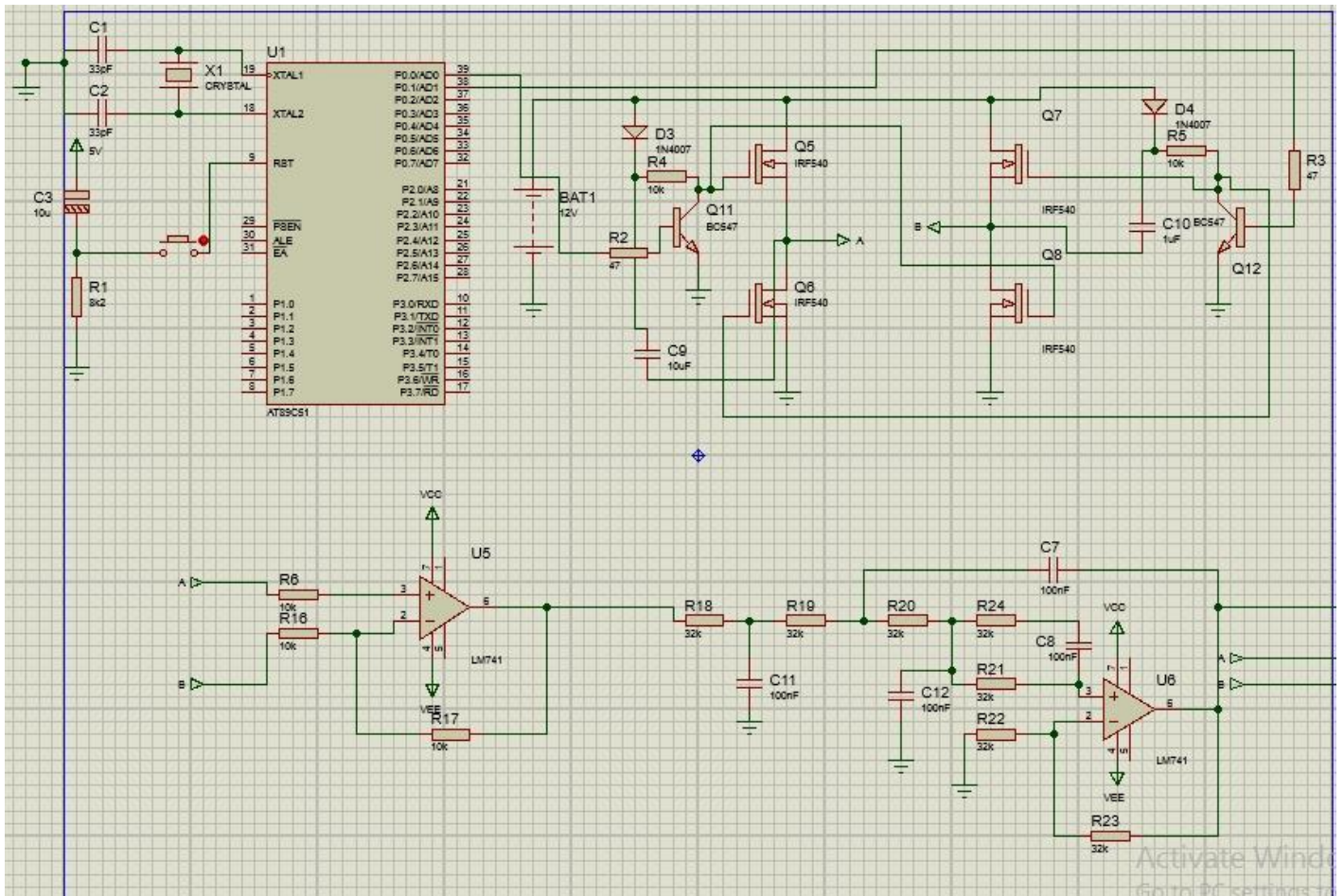
4. **Op-Amps (LM741):** The LM741 operational amplifiers are used in the subtractor and low-pass filter circuits. They combine the two square wave outputs from the MOSFETs and smooth the resulting signal.



5. **Capacitors and Resistors:** These passive components are used to create the low-pass filter and ensure stability and proper functionality of the circuits.
6. **Diodes:** Diodes are included in the circuit for flyback protection and to prevent reverse polarity that could damage the MOSFETs.
7. **12V DC Battery:** This battery serves as the primary power source for the inverter circuit, supplying the necessary voltage to the MOSFETs and other components.
8. **Software:** Proteus for simulation and KiCAD for schematic and PCB design.

3.3 Circuit Design

1. **8051 Microcontroller Setup:** The 8051 is programmed to generate a 50Hz square wave using its internal timers. Two complementary signals are outputted to control the gates of the two MOSFETs, ensuring one switches on while the other is off.
2. **MOSFET Switching:** The MOSFETs are connected to the 12V DC battery, and their switching action converts the DC power into two square wave signals. These signals are 180° out of phase, creating an inverted output on one of the MOSFETs.
3. **Subtractor Circuit:** The two complementary MOSFET outputs are fed into an op-amp subtractor circuit. This circuit cancels out the out-of-phase components, leaving behind a single 12V square wave signal.
4. **Low-Pass Filter:** The 12V square wave is then passed through a 4th-order low-pass filter. The filter smooths out the sharp edges of the square wave, transforming it into an 11V sine wave. This process removes high-frequency components, resulting in a clean AC waveform.



Circuit Implemented in Proteus Software

Chapter 4

Simulation And Results

The complete circuit was simulated using Proteus software to validate the design and confirm the correct functionality of the digital firing circuit for the single-phase inverter. The simulation environment included the following key steps :

1. Square Wave Generation:

- The simulation confirmed that the 8051 micro-controller successfully generated two complementary square waves at 50Hz. This output was essential for driving the MOSFETs.

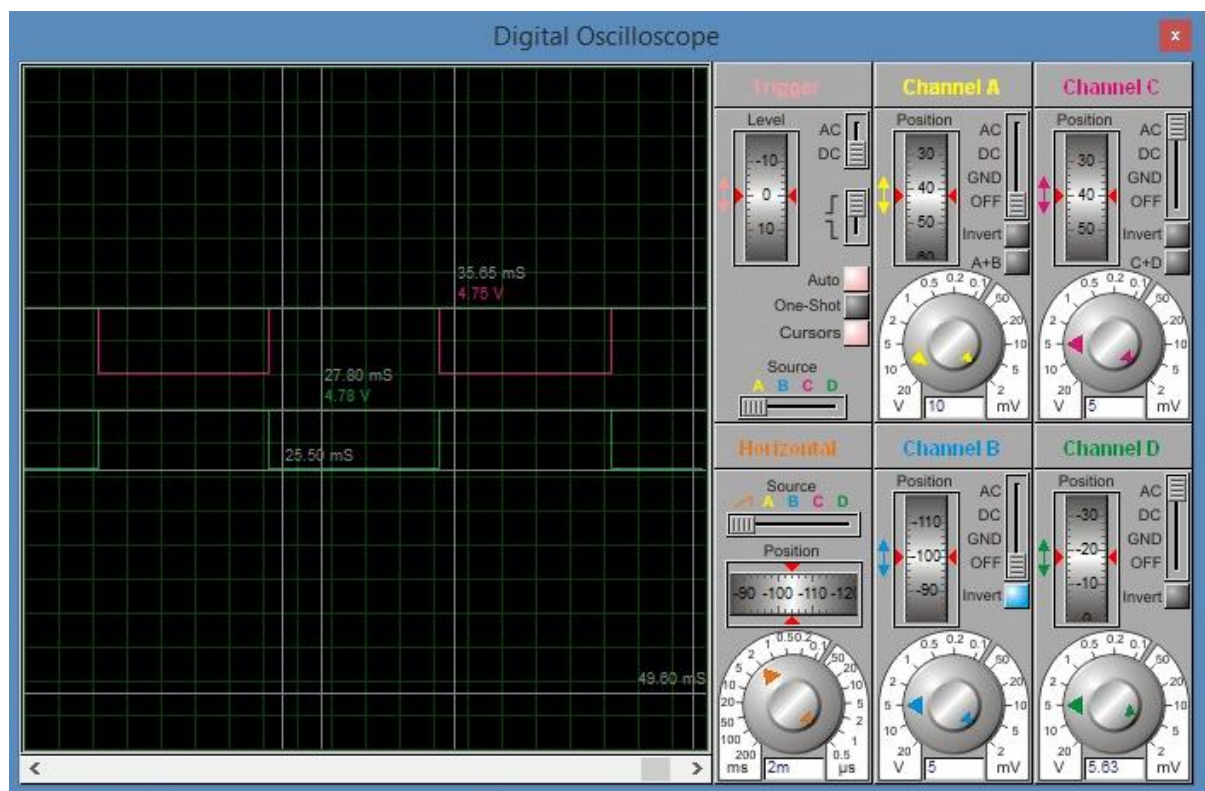


Figure 4.1 Image of The Square Wave Output

2. MOSFET Switching:

- The IRF540 MOSFETs switched on and off in response to the square wave signals. This operation was verified by observing the gate signals and the drain-source output. The output reflected the expected square wave behavior, demonstrating effective switching.

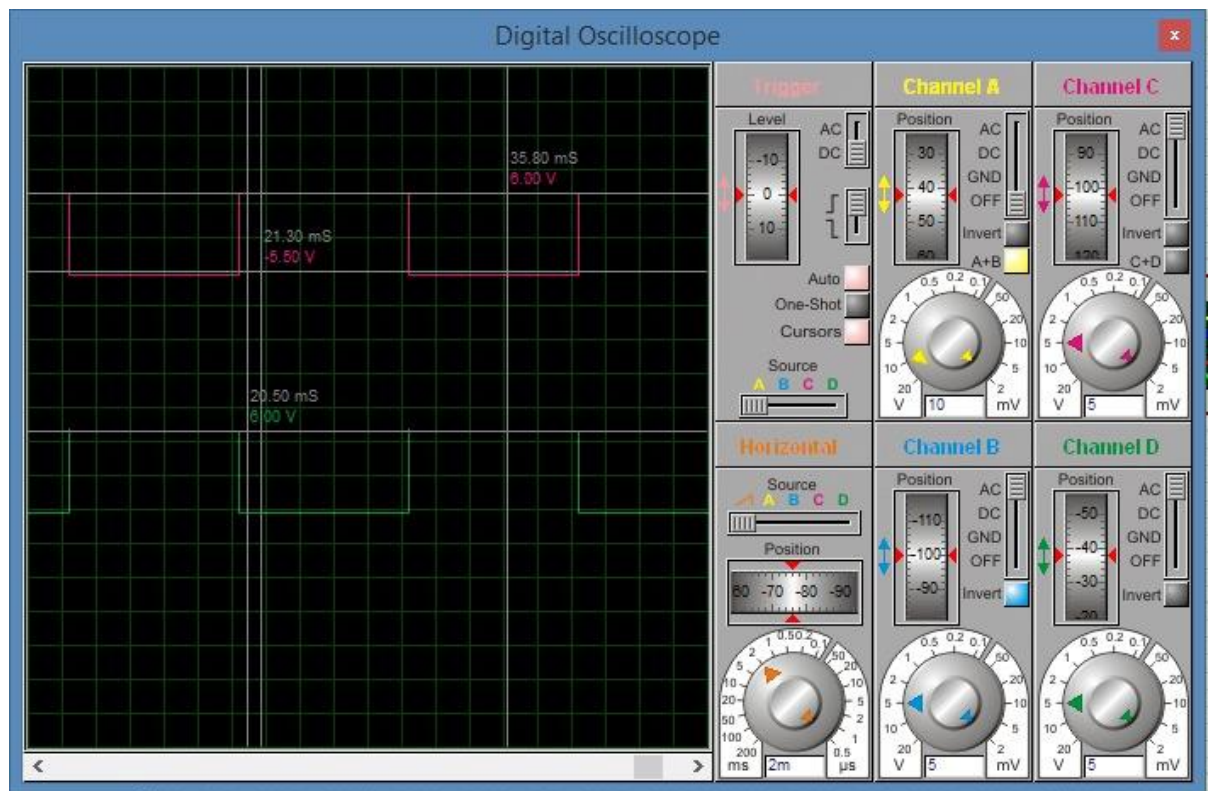


Figure 4.2 Image of The MOSFET Output

3. Subtractor Output:

- The combined output from the two MOSFETs was processed through the op-amp subtractor circuit, yielding a single 12V square wave. This result indicates that the subtractor circuit successfully canceled out the out-of-phase components.

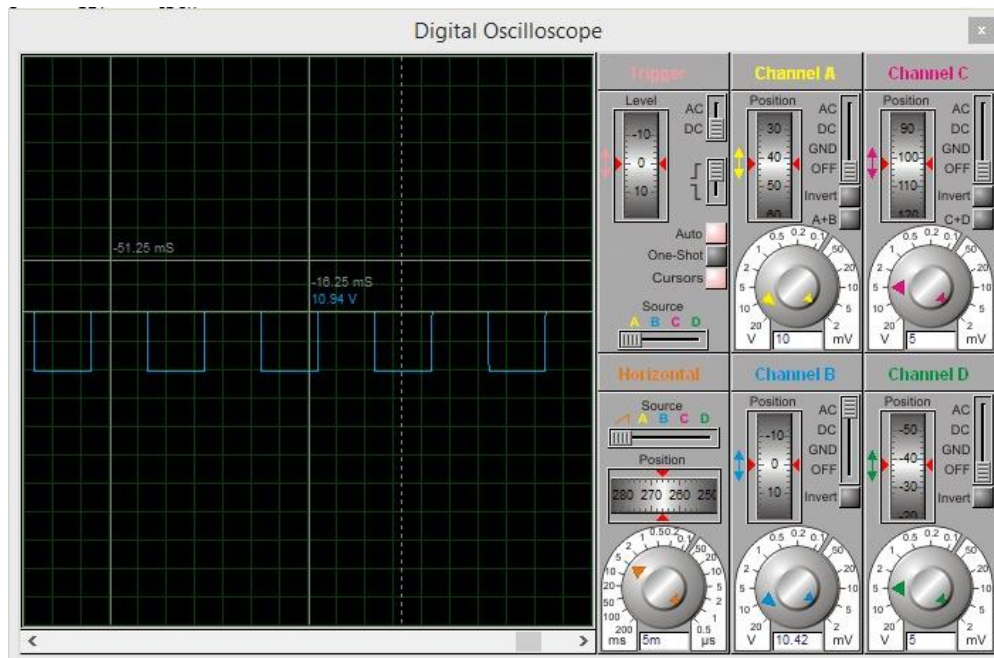


Figure 4.3 Image of The Subtractor Output

4. Low-Pass Filter Output:

- The 12V square wave from the subtractor was then passed through the low-pass filter. The simulation results showed that the output was transformed into an 11V sine wave, effectively smoothing the signal and reducing harmonic distortion.

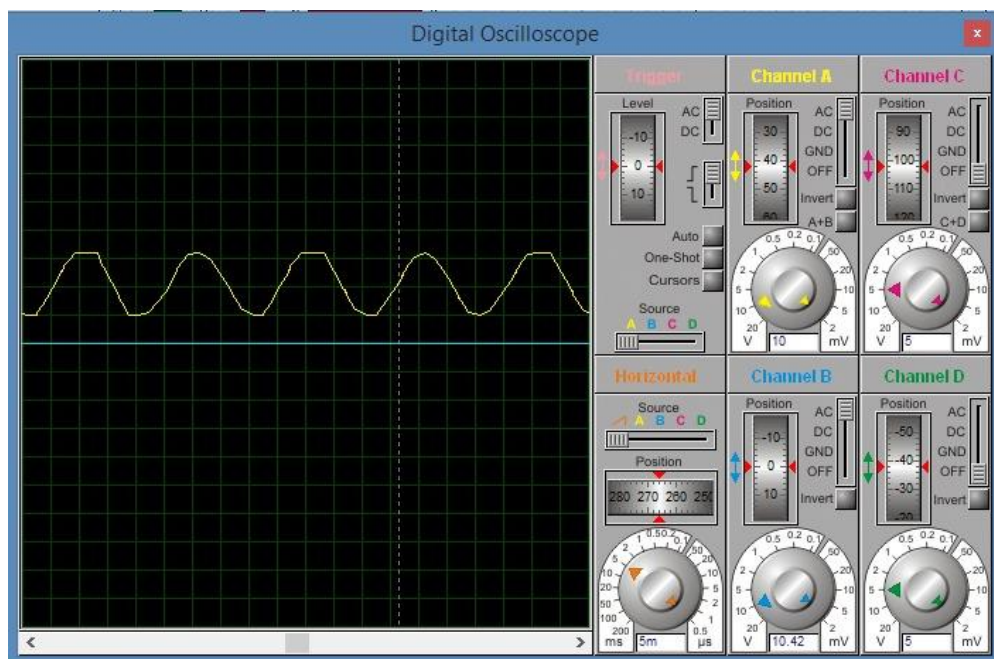


Figure 4.4 Image of The Low-Pass Filter Output

Output Waveform

The final output waveform, representing the converted AC signal, was obtained after passing through the low-pass filter. The simulation displayed a clean 11V sine wave, indicating that the circuit effectively converted the DC input into a usable AC output suitable for applications requiring a sine wave input.

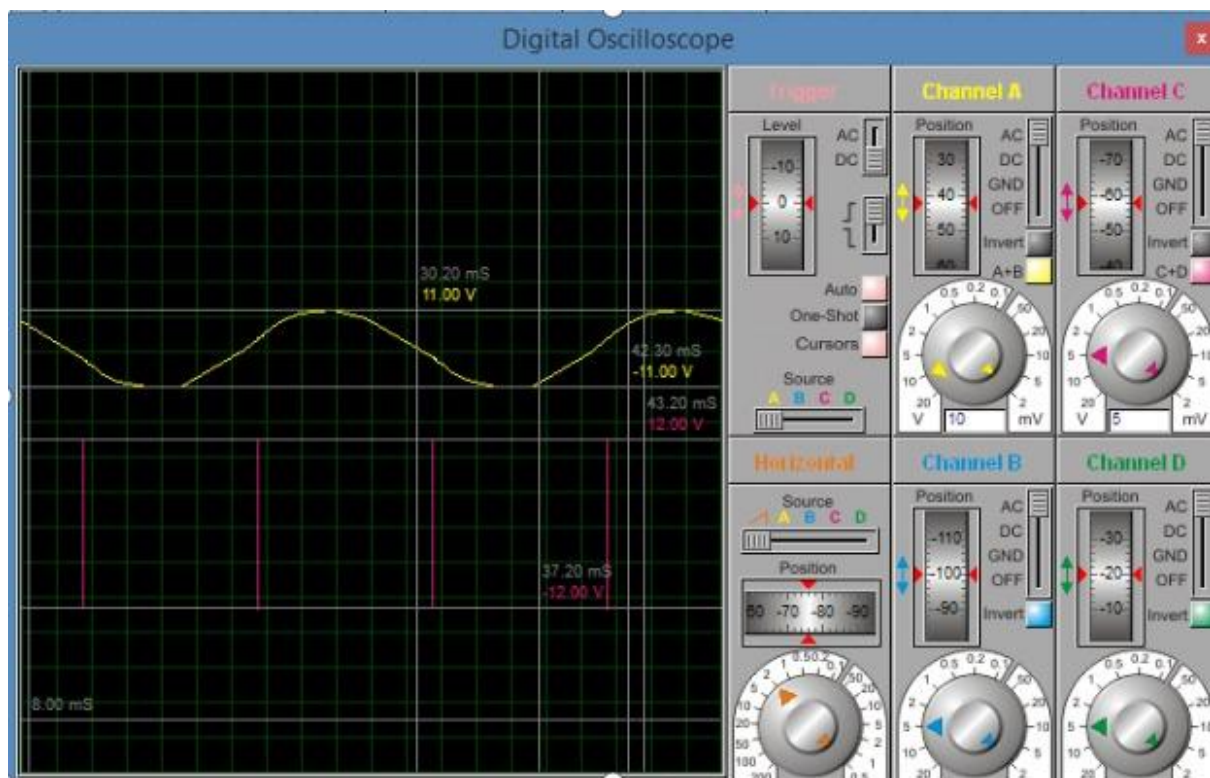


Figure 4.5 Image of The Sine wave Output

Discussion

The simulation results demonstrate that the design and implementation of the digital firing circuit for the single-phase inverter successfully met the project objectives. The key findings include:

- The 8051 micro-controller effectively generates the required square wave signals.
- The MOSFETs operate as intended, converting the DC supply into square wave outputs.
- The op-amp subtractor successfully processes the MOSFET outputs to create a single square wave.
- The low-pass filter effectively smooths the signal into an 11V sine wave with minimal distortion.

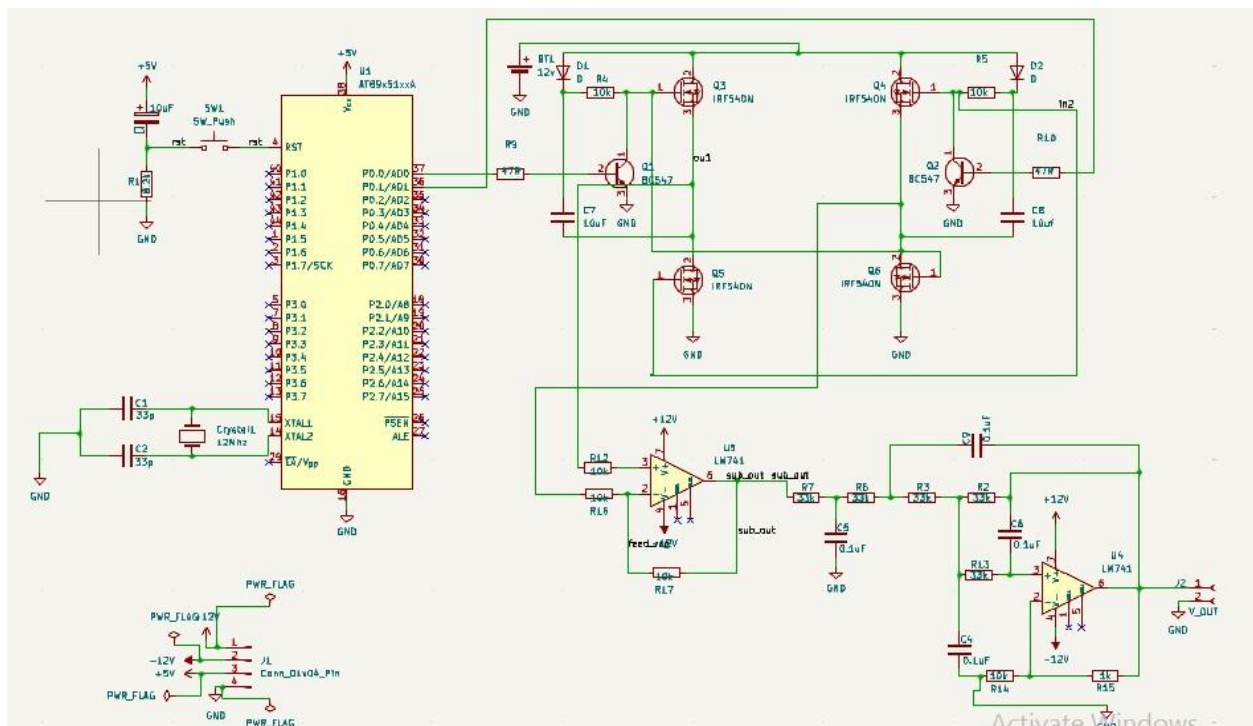
These results validate the effectiveness of the circuit design and confirm its readiness for further prototyping and physical implementation.

PCB Design

The PCB design process is a systematic approach to creating a functional and reliable printed circuit board. Below are the key steps involved in the design of the PCB for the digital firing circuit of a single-phase inverter using KiCAD.

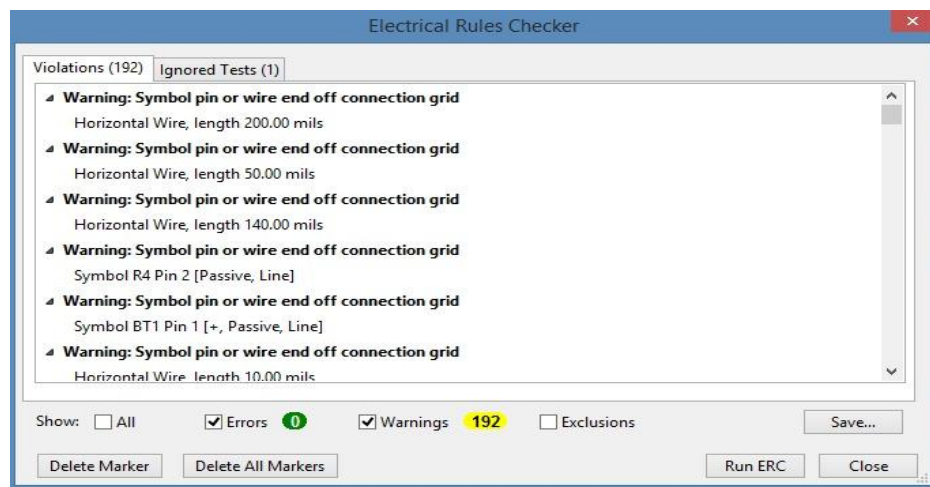
1. Schematic Capture:

- The first step in the PCB design process is creating the schematic diagram in KiCAD. This diagram serves as the foundation for the PCB layout.
- In this project, the schematic included the 8051 microcontroller, MOSFETs (IRF540), transistors (BC547), operational amplifiers (LM741), resistors, capacitors, diodes, and the 12V battery.
- Each component was added from the KiCAD library, ensuring that the correct symbols and footprints were used.



2. Electrical Rule Check (ERC):

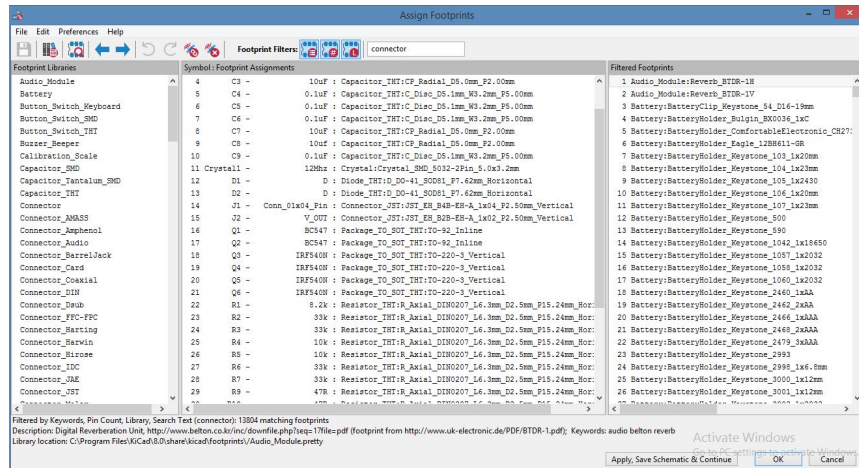
- After completing the schematic, an ERC was performed to identify any potential errors in the connections or component values. This step helps to catch mistakes early in the design process.
- The ERC checks for issues such as unconnected pins, incorrect component values, and other electrical inconsistencies that could affect circuit performance.



3. Assigning Footprints

After validating the schematic, the next step is to assign appropriate footprints to each component.

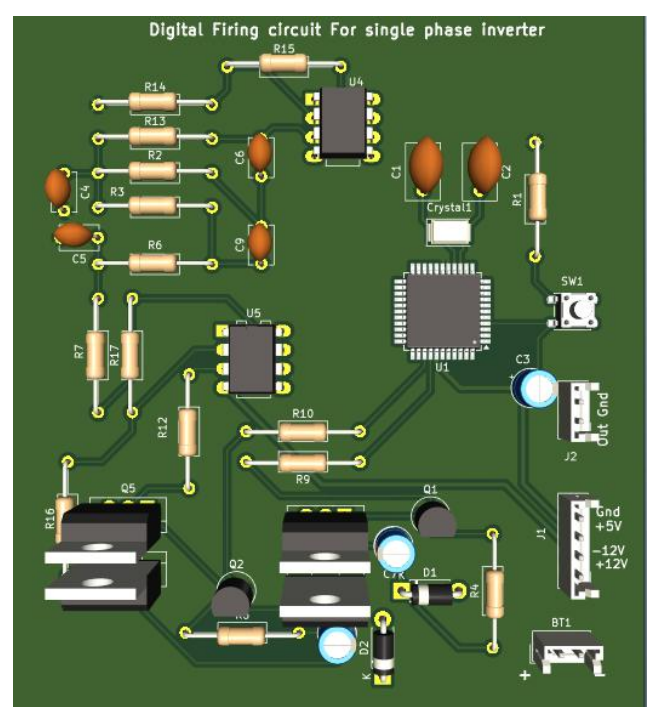
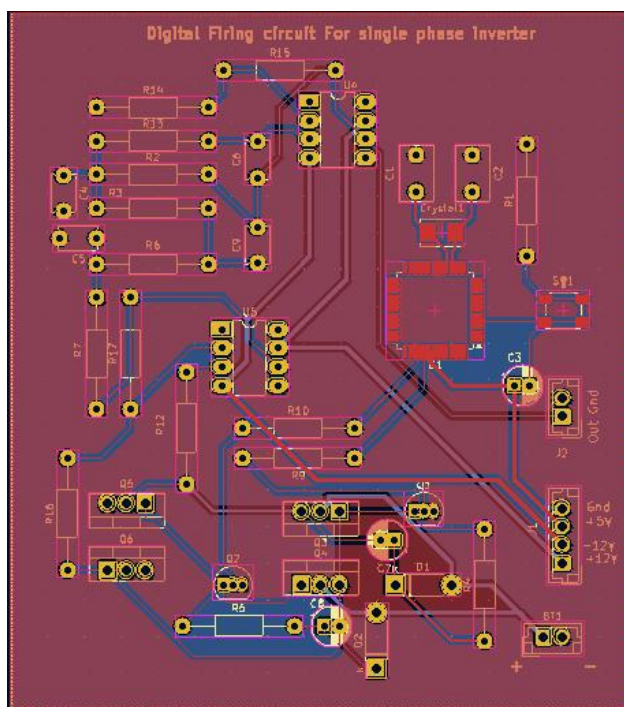
- In KiCAD, each component in the schematic is linked to a footprint that defines its physical dimensions and pin layout.
- Carefully selected footprints based on the size and design constraints of the PCB. For example, the 8051 microcontroller may require a specific DIP (Dual In-line Package) footprint, while the MOSFETs may require a TO-220 or similar footprint



In this image the footprints are assigned is shown

4. PCB Layout Design

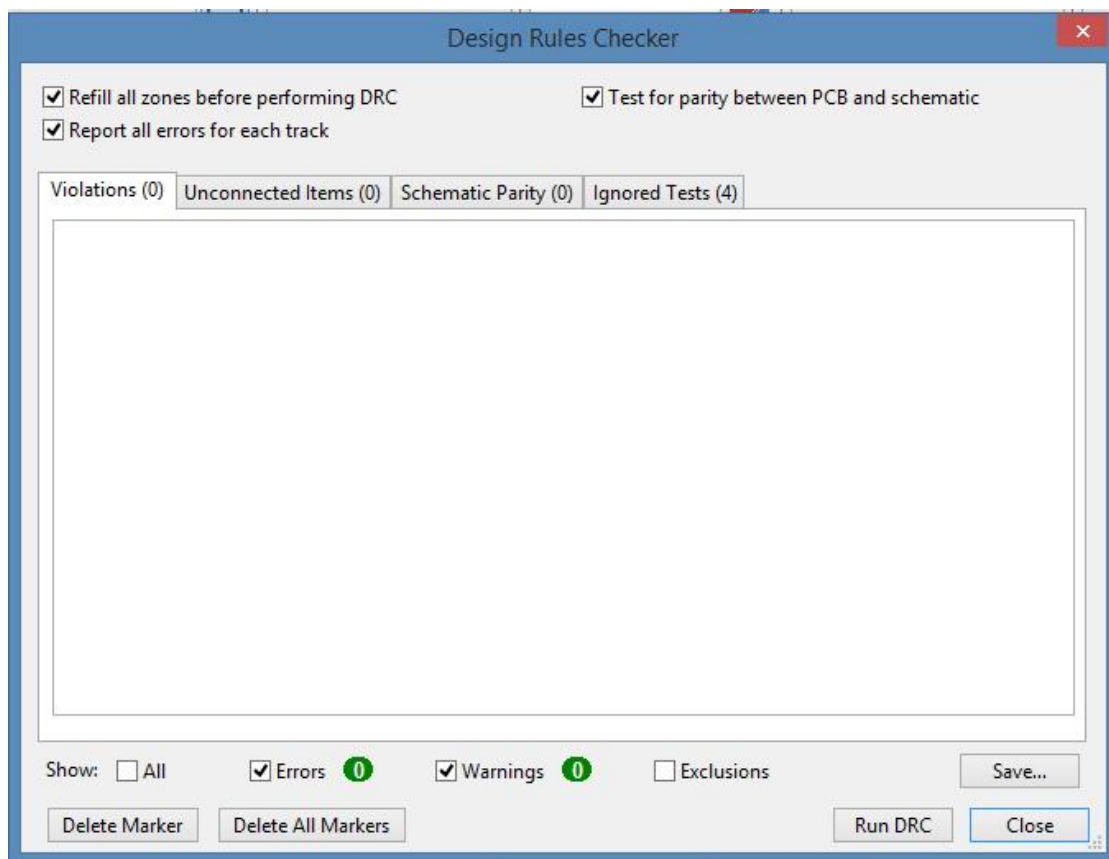
- Once the schematic is verified, the next step is to create the PCB layout. This involves placing components on a board and routing connections between them.
- The layout design must consider factors such as component placement, trace width, and routing techniques to minimize noise and ensure signal integrity.
- In this project, careful placement of components was crucial for achieving optimal performance, especially for the high-frequency signals from the op-amps and the switching MOSFETs.



The images show the PCB layout of the digital firing circuit and its 3D view, displaying component placement and final board appearance

5. DRC Check

- After completing the PCB layout, a Design Rule Check (DRC) is performed to ensure that the design adheres to manufacturing specifications.
- The DRC tool in KiCAD to analyze the layout for compliance with design rules, such as trace width, spacing, and clearance between components.
- Errors are reported by the DRC, ensuring that the design is manufacturable and reliable.
- The DRC check is crucial for avoiding potential manufacturing issues that could lead to circuit failures or performance problems.



Chapter 6

Conclusion

- This project successfully achieved the design and simulation of a digital firing circuit for a single-phase inverter using an 8051 microcontroller to generate complementary 50Hz square waves for triggering MOSFETs. The circuit, powered by a 12V battery, efficiently combined the MOSFET outputs using a subtractor circuit with an LM741 op-amp, and then passed the signal through a 4th-order low-pass filter to produce a clean 11V sine wave.
- Proteus was used to simulate the complete circuit, confirming its correct functionality. The PCB design was developed in KiCAD, starting with the schematic creation, followed by running ERC to validate the electrical design. Footprints were carefully assigned to ensure proper component placement, and the PCB was routed with signal integrity in mind. The DRC check ensured there were no layout violations, and the final design was ready for fabrication.
- The project successfully met its objectives by demonstrating effective DC-to-AC conversion with minimal distortion, using both digital and analog techniques. The results provide a solid foundation for further development and potential real-world application in inverter systems.

Reference

- 1) Textbook: The 8051 Microcontroller Based Embedded System, Manish k Patel
- 2) Textbook: Power Electronics By M.D.Singh
- 3) All component specifications and design parameters were referenced from datasheets available at Datasheets.com.

Appendix

Microcontroller Code:

```
$NOMOD51
$INCLUDE (8051.MCU)
ORG 0000H
LJMP MAIN ; interrupt vector is bypassed by main program
ORG 000BH ; ISR of Timer 0 at its vector address
CLR TR0 ; stop Timer 0
CPL P0.0 ; toggle P0.0 to generate square wave
CPL P0.1 ; toggle P0.1 to generate same square wave
MOV TH0, #0D8H ; reload count in Timer 0
MOV TL0, #0F0H
SETB TR0; start Timer 0 again
RETI ;return to the main program
ORG 001BH ;ISR of Timer 1 at its vector address
RETI ;return to the main program
ORG 0100H; main program starts at address 100H
MAIN: MOV TMOD, #01H ;Timer 0 Mode 0, Timer 1 not used
MOV IE, #82H ; enable Timer 0 interrupt
MOV TH0, #0D8H; load count in the timer registers
MOV TL0, #0F0H
SETB P0.0
CLR P0.1
SETB TR0 ; start Timer 0
HERE: SJMP HERE ; wait until Timer 0 overflows
END
```